

Driver & Car Booking System

Clear Plan & Process (Including Self-Drive)

TOTAL USER TYPES (VERY CLEAR)

1. Customer (User)

Can:

- Book **ride seats**
 - Book **full car**
 - Book **self-drive car**
 - Hire **driver**
 - Make payments
 - Give reviews
-

2. Car Owner

Can:

- Add cars
 - Set:
 - Ride sharing
 - Self-drive allowed
 - Driver required or not
 - Set price (per seat / per day)
-

3. Driver

Can:

- Register & upload license
- Set availability

- Accept bookings
 - Get paid
-

🔧 4. Admin

Can:

- Approve drivers & cars
- Verify documents
- Manage payments & refunds
- View reports

👉 **Only ONE User table**

Roles handled by role field

2 BOOKING TYPES (PROCESS CLEARLY)

◆ A. Ride Booking (Seat / Full Car)

User → Search Ride → Select Seats → Pay → Confirm

◆ B. Driver Hiring

User → Choose Driver → Select Date → Pay → Driver Assigned

◆ C. Self-Drive Car Booking ★

User → Choose Car → Upload License → Select Dates → Pay + Deposit → Admin Approves → Car Active

3 SELF-DRIVE BOOKING PROCESS (STEP-BY-STEP)

STEP 1: Car Owner Adds Car

- Car details
- Seats
- Price per day

- Allow self-drive = YES
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STEP 2: User Selects Self-Drive Car

- Choose start date & end date
 - System calculates:
 - Total days
 - Rental price
 - Security deposit
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STEP 3: User Uploads Documents

- Driving license
 - ID proof
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STEP 4: Payment

- Rental amount
 - Refundable deposit
-

STEP 5: Admin Approval

- Verify documents
 - Approve booking
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STEP 6: Booking Active

- Car is blocked
 - Status = ACTIVE
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STEP 7: Completion

- Car returned
 - Deposit refunded (if no damage)
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DATABASE FLOW (EASY TO UNDERSTAND)

MAIN TABLES

USER

└─ CAR

└─ RIDE

└─ BOOKING

└─ SELF_DRIVE_BOOKING

└─ PAYMENT

└─ REVIEW

5 API FLOW (SIMPLE)

Authentication

- Register
 - Login
 - Profile
-

Car APIs

- Add car
 - List cars
 - Enable self-drive
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Ride APIs

- Post ride
 - Search ride
 - Book seat
-

Self-Drive APIs

- List self-drive cars
 - Book self-drive
 - Upload license
 - Cancel booking
-

Payment APIs

- Create payment
 - Verify payment
 - Refund
-

Admin APIs

- Approve drivers
 - Approve self-drive booking
 - View reports
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DEVELOPMENT PLAN (DAY-WISE)

Week 1 – Setup

- Project setup
- Database design
- Git setup
- User roles

📅 Week 2 – Core Features

- Auth (JWT)
- Car management
- Ride posting
- Ride search

📅 Week 3 – Business Logic

- Seat calculation
- Driver booking
- Self-drive logic
- Availability checks

📅 Week 4 – Payments

- Razorpay/Stripe
- Deposit handling
- Refunds
- Invoices

📅 Week 5 – Admin & Security

- Admin dashboards
- Document verification
- Permissions
- Reports

📅 Week 6 – Testing & Deployment

- API testing
- Bug fixes
- AWS deployment
- Final launch